

## Java Programming Lab (Control Statements)

**Assignment 1:** Write a Java program that accepts a 5-digit number and prints:

- Sum of digits at even positions
- Sum of digits at odd positions
- Difference between the two

Example:

Input: 54321

Odd position sum =  $5+3+1$

Even position sum =  $4+2$

**Assignment 2:** Write a Java program that classifies a number as:

- Positive / Negative / Zero
- Even / Odd
- Multiple of 5 or not
- Three-digit or not

Display all classifications.

**Assignment 3:** Write a Java program to compute salary using:

- Basic salary input
- HRA = 20% of basic if  $\text{basic} \leq 20000$  otherwise 30%
- DA = 80% of basic
- Tax = 10% if gross salary > 50000

Display gross and net salary.

**Assignment 4:**

Write a Java program to check whether a two-digit number is **special**.

A number is special if:

Sum of digits + Product of digits = original number

Example:

$59 \rightarrow 5+9 + 5 \times 9 = 14 + 45 = 59$

**Assignment 5:** Write a Java program to compute:

$1 - 2 + 3 - 4 + 5 - 6 \dots N$  terms

Display the final sum.

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**Assignment 6:** Write a Java program that counts how many times each digit appears in a number.

Example:

Input: 1223401

Output:

1 → 2 times

2 → 2 times

3 → 1 time

4 → 1 time

0 → 1 time

### Assignment 7

Write a Java program to print:

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1
212
32123
4321234
```

Use nested loops.

**Assignment 8:** Write a Java program that counts how many digits of a number **divide the number exactly**.

Example:

Input: 124

Digits: 1,2,4

All divide 124 → count = 3

**Assignment 9:** Write a Java program to generate the series:

2, 6, 7, 21, 22, 66, 67, ...

Pattern:

×3, +1, ×3, +1 ...

Print first N terms.

**Assignment 10:** Write a Java program that:

- Replaces all 0 with 1
- Replaces all 1 with 0
- Keeps other digits same

Example:

Input: 102301

Output: 012310